

Journal 10: Reflecting on Key Learning and Future Applications

Sharie Rhea

Southern New Hampshire University

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Brian Enochson

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Throughout this course I encountered a few challenges. For a few of the programming assignments, I attempted to implement functionality without fully reading the associated documentation beforehand. Naturally, this led to code that was full of logical errors, such as the central processing unit (CPU) percentage always being zero on the first read from the psutil library if an interval is not specified. I overcame this challenge by reviewing the documentation more thoroughly, and will remember to continue this best practice in the future. Another challenge that I overcame was a lack of familiarity with various operating systems. Despite using computers for much of my life, I have really only been exposed to a few Linux distributions. Throughout this course I learned so much more about Windows and Mac, and explored Windows a bit more fully on my own.

I plan to apply the knowledge I've learned in my remaining coursework here at Southern New Hampshire University. I still have eight more classes to obtain my master's, and I am certain that proper programming practices will enhance my future assignments. Additionally, a deeper understanding of security, access control, and optimization will help me to write better software as a research computer scientist and strengthen my career. In fact, I was surprised at how much I learned surrounding virtual memory and issues such as cache coherency. One task I currently have at work is to implement shared memory between two microcontroller cores, and my understanding of these concepts has been invaluable.

I applied operating system principles and best practices in my programming assignments by following coding standards and guidelines as well as properly handling errors. Additionally, I considered process management, security, and optimization when creating programs, ensuring that code was efficient and minimal while still accomplishing the assigned task. For written

assignments, I applied my research and learning to explain and apply concepts such as processing scheduling, memory management, multithreading, security, input/output (I/O) devices, and file systems.